

Qiao Duan

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Education

University of Cambridge, Ph.D in Physics October 2024 - October 2028
3rd Year PhD student working on Extragalactic Astronomy and Early Universe Research using JWST data. Research focuses on Galaxy Stellar Populations, and Star Formation Properties, Spectral Energy Distribution (SED) fitting, and Galaxy Mergers.

The University of Manchester, MPhys(Hons) Physics with Astrophysics September 2020 - June 2024
Astrophysics Student with Courses in: Galaxies, Stars and Stellar Evolution, Exoplanets, Gravitation, Early Universe, Cosmology, Quantum Mechanics, Electrodynamics, Optics, Statistical Mechanics, Nuclear and Particle Physics, Nonlinear Physics
First Class Honour Degree

Talks

Contributed Talk — <i>Kavli Astrophysics Symposium: Forging the Frontier of Cosmic Discovery</i>	September 2026
Contributed Talk — <i>Galaxies and the 21-cm Frontier: Unveiling the Epoch of Reionization</i>	May 2026
Contributed Talk — <i>Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration</i>	March 2026
Contributed Talk — <i>KICC Science Focus Meeting: Towards the Future of SED Fitting</i>	July 2025
Contributed Talk — <i>JADES ESA Meeting</i>	June 2025
Contributed Talk — <i>Galaxy Origins in the JWST Era: A “Crisol” of Stars, ISM, and Super-massive Black Holes in the City of the Three Cultures</i>	May 2025

Observing programs

JWST Cycle 5 GO 10790: *LAPIS: Large Area Parallel Imaging Survey* – 1000h — Co-Investigator

Mentorship

Summer Project — <i>LRDs as Direct-Collapse Black Holes</i>	(4 months), June 2026–Present
Master’s Project — <i>Clumpy Galaxy Fraction across Cosmic Time</i>	(7 months), December 2025–July 2026

Awards

Trinity College Scholarship — £198,000
ESA JADES Award — €650

Professional service

AAS Referee	Since July 2026
LOC — <i>Kavli Astrophysics Symposium: Forging the Frontier of Cosmic Discovery</i>	September 2026
LOC — <i>BeyondHubble: Revisiting the Hubble Sequence Across Cosmic Time</i>	April 2026

Referees

Prof. Sandro Tacchella
st578@cam.ac.uk

Prof. Roberto Maiolino
rm665@cam.ac.uk

Prof. Christopher Conselice
conselice@manchester.ac.uk

- 1) *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: stellar population catalogue for galaxies in GOODS-N and GOODS-S*
Qiao Duan, Sandro Tacchella, Benjamin D. Johnson, Brant Robertson, *et al.*
Status: Preprint [\[ADS\]](#) **Citations:** 3
- 2) *Galaxy mergers in the epoch of reionization – I. A JWST study of pair fractions, merger rates, and stellar mass accretion rates at $z = 4.5\text{--}11.5$*
Qiao Duan, Christopher J. Conselice, Qiong Li, Duncan Austin, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 89
- 3) *Galaxy mergers in the epoch of reionization – II. Merger-triggered star formation and AGN activities at $z = 4.5\text{--}8.5$*
Qiao Duan, Christopher J. Conselice, Thomas Harvey, Qiong Li, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 24
- 4) *Adding value to JWST spectra and photometry: stellar population and star formation properties of spectroscopically confirmed JADES and CEERS galaxies at $z > 7$*
Qiao Duan, Christopher J. Conselice, Qiong Li, Thomas Harvey, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 19

Lead-author publications

Citations: 35

- 1) *JADES: the mass–metallicity relation at $z = 1\text{--}10$. New calibrations, extremely metal-poor galaxies, and chemical diversity*
Yuki Isobe, Mirko Curti, Roberto Maiolino, Qiao Duan, *et al.*
Status: Submitted to MNRAS [\[ADS\]](#) **Citations:** 9
- 2) *Metal Mayhem at $z \sim 7\text{--}10$: Diversity and Evolution of Gas-Phase Metallicity Gradients*
Maria Koller, Roberto Maiolino, Hannah Übler, Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 3
- 3) *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCIm Imaging in GOODS-S and GOODS-N*
Benjamin D. Johnson, Brant E. Robertson, Daniel J. Eisenstein, ..., Qiao Duan, *et al.*
Status: Submitted to AAS Journals [\[ADS\]](#) **Citations:** 21
- 4) *A dual AGN at $z = 5.4$ associated with a Lyman- α Nebula in the Center of a Cosmic Filament*
Qiong Li, Christopher J. Conselice, Qiao Duan, Duncan Austin, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 2

Co-authored publications

Citations: 610

- 1) *JADES: Dynamical Measurements of Dark Matter Halo Masses at $z \approx 7$ Using Galaxy Pairs*
Zihao Wu, Daniel J. Eisenstein, Benjamin D. Johnson, ..., Qiao Duan, *et al.*
Status: Preprint [\[ADS\]](#) **Citations:** 0
- 2) *The Roman eXtreme Deep Field (RXDF)*
Haojing Yan, Anton M. Koekemoer, Yue Shen, ..., Qiao Duan, *et al.*
Status: Published in the Open Journal of Astrophysics [\[ADS\]](#) **Citations:** 0
- 3) *A Steep-Extinction Quasi-stellar Object at $z = 4.6$: JWST Evidence for Abundant Small Dust Grains*
Mingyu Li, Zheng Cai, Roberto Maiolino, ..., Qiao Duan, *et al.*
Status: Published in ApJL [\[ADS\]](#) **Citations:** 0
- 4) *GA-NIFS: Dissecting The Alchemised: JWST reveals turbulent metal-poor gas fuelling a co-spatial starburst in a complex system at $z = 10.17$*
Robert G. Pascalau, Francesco D'Eugenio, Roberto Maiolino, Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 4

- 5) *Clumps in High-Redshift Galaxies: Mass Scaling and Radial Trends from JADES*
Yongda Zhu, Marcia J. Rieke, Zhiyuan Ji, . . . , Qiao Duan, *et al.*
Status: Published in ApJ [\[ADS\]](#) **Citations:** 6
- 6) *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometrically Selected Galaxy Candidates at $z > 8$*
Kevin N. Hainline, Daniel J. Eisenstein, Lily Whitler, . . . , Qiao Duan, *et al.*
Status: Published in ApJ [\[ADS\]](#) **Citations:** 16
- 7) *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Catalogs of inferred morphological properties of galaxies from JWST/NIRCam imaging in GOODS-N and GOODS-S*
Courtney Carreira, Brant E. Robertson, A. Lola Danhaive, . . . , Qiao Duan, *et al.*
Status: Submitted to AAS Journals [\[ADS\]](#) **Citations:** 12
- 8) *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog*
Brant E. Robertson, Benjamin D. Johnson, Sandro Tacchella, . . . , Qiao Duan, *et al.*
Status: Submitted to AAS Journals [\[ADS\]](#) **Citations:** 24
- 9) *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: MIRI Coordinated Parallels in GOODS-S and GOODS-N*
Stacey Alberts, Daniel J. Eisenstein, Andrew J. Bunker, . . . , Qiao Duan, *et al.*
Status: Preprint [\[ADS\]](#) **Citations:** 4
- 10) *The dark side of early galaxies: geko uncovers dark-matter fractions at $z \sim 4-6$*
A. Lola Danhaive, Sandro Tacchella, Andrew J. Bunker, . . . , Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 9
- 11) *Mergers lighting the early Universe: modest star-formation enhancement without evidence for AGN or Ly α triggering*
Dávid Puskás, Sandro Tacchella, Charlotte Simmonds, . . . , Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 10
- 12) *JADES Data Release 4 – Paper II: Data reduction, analysis and emission-line fluxes of the complete spectroscopic sample*
Jan Scholtz, Stefano Carniani, Eleonora Parlanti, . . . , Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 48
- 13) *JADES Data Release 4 Paper I: Sample Selection, Observing Strategy and Redshifts of the complete spectroscopic sample*
Emma Curtis-Lake, Alex J. Cameron, Andrew J. Bunker, . . . , Qiao Duan
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 42
- 14) *Bursting at the seams: the star-forming main sequence and its scatter at $z = 3-9$ using NIRCam photometry from JADES*
Charlotte Simmonds, Sandro Tacchella, William McClymont, . . . , Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 58
- 15) *Lonely Little Red Dots: Challenges to the AGN-nature of little red dots through their clustering and spectral energy distributions*
María Carranza-Escudero, Christopher J. Conselice, Nathan Adams, . . . , Qiao Duan, *et al.*
Status: Published in ApJL [\[ADS\]](#) **Citations:** 25
- 16) *When relics were made: vigorous stellar rotation and low dark matter content in the massive ultra-compact galaxy GS-9209 at $z = 4.66$*
Robert G. Pascalau, Francesco D'Eugenio, Sandro Tacchella, . . . , Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 18
- 17) *JADES and SAPPHIRES: Galaxy Metamorphosis Amidst a Huge, Luminous Emission-line Region*
Francesco D'Eugenio, Jakob M. Helton, Kevin N. Hainline, . . . , Qiao Duan, *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 9

- 18) *Constraining the major merger history of $z \sim 3\text{--}9$ galaxies using JADES: dominant in-situ star formation*
 Dávid Puskás, Sandro Tacchella, Charlotte Simmonds, ..., [Qiao Duan](#), *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 56
- 19) *EPOCHS XI: The Structure and Morphology of Galaxies in the Epoch of Reionization to $z \sim 12.5$*
 Lewi Westcott, Christopher J. Conselice, Thomas Harvey, ..., [Qiao Duan](#), *et al.*
Status: Published in ApJ [\[ADS\]](#) **Citations:** 17
- 20) *EPOCHS I. The Discovery and Star Forming Properties of Galaxies in the Epoch of Reionization at $6.5 < z < 18$ with PEARLS and Public JWST data*
 Christopher J. Conselice, Nathan Adams, Thomas Harvey, ..., [Qiao Duan](#), *et al.*
Status: Published in ApJ [\[ADS\]](#) **Citations:** 44
- 21) *EPOCHS Paper X: Environmental effects on Galaxy Formation and Protocluster Galaxy candidates at $4.5 < z < 10$ from JWST observations*
 Qiong Li, Christopher J. Conselice, Florian Sarron, ..., [Qiao Duan](#), *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 29
- 22) *EPOCHS III: Unbiased UV continuum slopes at $6.5 < z < 13$ from combined PEARLS GTO and public JWST NIRCам imaging*
 Duncan Austin, Christopher J. Conselice, Nathan J. Adams, ..., [Qiao Duan](#), *et al.*
Status: Published in ApJ [\[ADS\]](#) **Citations:** 68
- 23) *Dust Extinction Measures for $z \sim 8$ Galaxies using Machine Learning on JWST Imaging*
 Kwan Lin Kristy Fu, Christopher J. Conselice, Leonardo Ferreira, ..., [Qiao Duan](#), *et al.*
Status: Published in MNRAS [\[ADS\]](#) **Citations:** 4
- 24) *EPOCHS IV: SED Modelling Assumptions and their impact on the Stellar Mass Function at $6.5 < z < 13.5$ using PEARLS and public JWST observations*
 Thomas Harvey, Christopher J. Conselice, Nathan J. Adams, ..., [Qiao Duan](#), *et al.*
Status: Published in ApJ [\[ADS\]](#) **Citations:** 107